

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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QUESTIONS

Q1

Banking Transaction System –
Exception Handling
10 marks

Q2

Hospital Patient Registration –
User-Defined Exception
10 marks

Q3

E-Commerce Order Processing
– Built-in Exceptions
10 marks

Q4

Hospital Emergency System –
Thread Priority
10 marks

Q5

Online Food Delivery – Multiple
Threads and Execution Order
10 marks

5/5 answered

Q5

Online Food Delivery – Multiple Threads
and Execution Order10
marks

An online food-delivery application uses three threads: Order Processing, Payment Processing, and Delivery Assignment.

Develop a Java program in which these activities are represented by separate threads. Analyze the problem if all three threads are started independently.

Modify the program so that:

- Order processing completes before payment processing.
- Payment processing completes before delivery assignment.
- The program uses appropriate Java thread-management techniques.
- The output clearly shows the required execution order.

Explain the difference between simply starting all threads and explicitly controlling their execution order.

Sample Test Cases:

Input / Scenario Expected Result Analysis Focus

Order=Pizza; Payment=UPI; Delivery=Agent A
Order → Payment → Delivery Normal execution

Order=Burger; Payment=Card; Delivery=Agent B

Order → Payment → Delivery Concurrency / design
Payment=Failed Delivery assignment should not
proceed Exception / boundary

Your Code

```
1 class Main
2 {
3     static void order()
4     {
5         System.out.println("Order Process:");
6     }
7     static void payment()
8     {
9         System.out.println("Payment Process:");
10    }
11    static void delivery()
12    {
13        System.out.println("Delivery Assignment:");
14    }
15    public static void main(String[] args) {
16        {
17            Thread t1=new Thread(()->order());
18            Thread t2=new Thread(()->payment());
19            Thread t3=new Thread(()->delivery());
20            t1.start();
21            t1.join();
22            t2.start();
23            t2.join();
24            t3.start();
25            t3.join();
26        }
27    }
```

Input

stdin input

Output

```
Order Processing  
Payment Processing  
Delivery Assignment
```

✓ Submitted

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